OPEN LOCUS OF NEGATIVE EXT VANISHING FOR A PSEUDO-COHERENT COMPLEX AGAINST A RELATIVELY PERFECT ONE, WITH AFFINE FINITELY PRESENTED HOM SPACE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file quot.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- Definition spaces-more-morphisms-definition-relatively-perfect (canonical tag 0DKN). cited at statement lines 5159-5160. E is Y-perfect.
- Lemma sites-cohomology-lemma-pseudo-coherent-pullback (canonical tag 08H4). cited at proof lines 5185-5187. K_V stays pseudo-coherent under base change.
- Lemma spaces-more-morphisms-lemma-base-change-relatively-perfect (canonical tag 0DKT). cited at proof lines 5188-5189. E_V stays V-perfect under base change.
- Lemma perfect-lemma-affine-morphism-and-hom-out-of-perfect (canonical tag 0AA7). cited at proof lines 5193-5198. Ext over fibres is stable under field extension.
- Lemma spaces-more-morphisms-lemma-compute-ext-rel-perfect (canonical tag 0DKY). cited at proof lines 5204-5206. L universally computes Rf_*RSheafHom(K, E) over affine Y.
- Lemma perfect-lemma-jump-loci (canonical tag 0BDI). cited at proof line 5221. The vanishing locus of H^i of a pseudo-coherent complex on fibres is open.
- Lemma sites-cohomology-lemma-section-RHom-over-U (canonical tag 08JA). cited at proof lines 5245-5246. Ext^i as H^i(X_V, RSheafHom).
- Lemma sites-cohomology-lemma-Leray-unbounded (canonical tag 0D6H). cited at proof lines 5246-5247. H^i(X_V, -) = H^i(V, Rf_{V,*}-).
- Lemma derived-lemma-negative-vanishing (canonical tag 05TC). cited at proof lines 5249-5250. Vanishing cohomology sheaves in degrees < 0 give vanishing H^i, $i < 0$.
- Lemma more-algebra-lemma-lift-pseudo-coherent-from-residue-field (canonical tag 0BCC). cited at proof lines 5260-5261. Fibre-wise vanishing lifts to stalks for pseudo-coherent complexes.

Date: stacks-project:quot.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

- Lemma spaces-simplicial-lemma-fppf-neg-ext-zero-hom (canonical tag 0DL9). cited at proof lines 5274 and 5283 (footnote). F is an fppf sheaf via the Cech hypercovering argument in the footnote.
- Lemma bootstrap-lemma-locally-algebraic-space-finite-type (canonical tag 04U0). cited at proof line 5287. Representability is etale local.
- Lemma spaces-morphisms-lemma-affine-local (canonical tag 03WG). cited at proof line 5289. Affineness is etale local.
- Lemma spaces-morphisms-lemma-finite-presentation-local (canonical tag 0410). cited at proof lines 5289-5290. Finite presentation is etale local.
- Lemma more-algebra-lemma-pseudo-coherent (canonical tag 064U). cited at proof lines 5300-5301. L is represented by a two-term complex of finite free modules in degrees 1, 0.

Source reference. Original definition 0DKN: sources/stacks-chow/tags.txt, source lines 17018–17018. Cited at statement lines 5159-5160: E is Y-perfect per this definition; load-bearing.

Source reference. Original lemma 08H4: sources/stacks-chow/tags.txt, source lines 10767–10767. Cited at proof lines 5185-5187: K_V is pseudo-coherent for any V -> Y; load-bearing (first observation).

Source reference. Original lemma 0DKT: sources/stacks-chow/tags.txt, source lines 17023–17023. Cited at proof lines 5188-5189: E_V is V-perfect; load-bearing (first observation).

Source reference. Original lemma 0AA7: sources/stacks-chow/tags.txt, source lines 12975–12975. Cited at proof lines 5197-5198: Ext^i over the fibre tensor k' equals Ext^i over k; load-bearing (well-definedness on |Y|).

Source reference. Original lemma 0DKY: sources/stacks-chow/tags.txt, source lines 17028–17028. Cited at proof lines 5204-5206: supplies L universally computing Rf_*RSheafHom(K, E) over affine Y; load-bearing.

Source reference. Original lemma 0BDI: sources/stacks-chow/tags.txt, source lines 14316–14316. Cited at proof line 5221: the locus where H^i(L x^L kappa(y)) = 0 for i > 0 is open; load-bearing for part (1).

Source reference. Original lemma 08JA: sources/stacks-chow/tags.txt, source lines 10843–10843. Cited at proof lines 5245-5246: first equality expressing Ext^i over X_V as H^i(X_V, RSheafHom); load-bearing for part (2).

Source reference. Original lemma 0D6H: sources/stacks-chow/tags.txt, source lines 16522–16522. Cited at proof lines 5246-5247: second equality H^i(X_V, -) = H^i(V, Rf_{V,*}-); load-bearing for part (2).

Source reference. Original lemma 05TC: sources/stacks-chow/tags.txt, source lines 7476–7476. Cited at proof lines 5249-5250: vanishing cohomology sheaves in degrees < 0 kills H^i for i < 0; load-bearing for part (2).

Source reference. Original lemma 0BCC: sources/stacks-chow/tags.txt, source lines 14275–14275. Cited at proof lines 5260-5261: W = Y gives H^i(L) = 0 for i > 0 on stalks; load-bearing for the claim.

Source reference. Original lemma 0DL9: sources/stacks-chow/tags.txt, source lines 17039–17039. Cited at proof lines 5274 and 5283 (including the footnote hypercovering argument): F is an fppf sheaf; load-bearing for part (3).

Source reference. Original lemma 04U0: sources/stacks-chow/tags.txt, source lines 6273–6273. Cited at proof line 5287: checking representability etale locally on Y; load-bearing for part (3).

Source reference. Original lemma 03WG: sources/stacks-chow/tags.txt, source lines 5111–5111. Cited at proof line 5289: affineness of F -> Y checked etale locally; load-bearing for part (3).

Source reference. Original lemma 0410: sources/stacks-chow/tags.txt, source lines 5275–5275. Cited at proof lines 5289-5290: finite presentation of F -> Y checked etale locally; load-bearing for part (3).

Source reference. Original lemma 064U: sources/stacks-chow/tags.txt, source lines 7878–7878. Cited at proof lines 5300-5301: L represented by O^m_1 -> O^m_0 in degrees 1, 0; load-bearing for the fibre-product presentation of F.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces over S. Assume f is proper, flat, and of finite presentation. Let $K, E \in D(\mathcal{O}_X)$. Assume K is pseudo-coherent and E is Y-perfect (More on Morphisms of Spaces, Definition 0DKN). For a field k and a morphism $y : \mathrm{Spec}(k) \to Y$ denote K_y, E_y the pullback to the fibre X_y.*

(1) *There is an open $W \subset Y$ characterized by the property*
$$y \in |W| \Leftrightarrow \mathrm{Ext}^i_{\mathcal{O}_{X_y}}(K_y, E_y) = 0 \text{ for } i < 0.$$
(2) *For any morphism $V \to Y$ factoring through W we have*
$$\mathrm{Ext}^i_{\mathcal{O}_{X_V}}(K_V, E_V) = 0 \quad \text{for} \quad i < 0$$
where X_V is the base change of X and K_V and E_V are the derived pullbacks of K and E to X_V.
(3) *The functor $V \mapsto \mathrm{Hom}_{\mathcal{O}_{X_V}}(K_V, E_V)$ is a sheaf on $(Spaces/W)_{fppf}$ representable by an algebraic space affine and of finite presentation over W.*

Proof. For any morphism $V \to Y$ the complex K_V is pseudo-coherent (Cohomology on Sites, Lemma 08H4) and E_V is V-perfect (More on Morphisms of Spaces, Lemma 0DKT). Another observation is that given $y : \mathrm{Spec}(k) \to Y$ and a field extension k'/k with $y' : \mathrm{Spec}(k') \to Y$ the induced morphism, we have
$$\mathrm{Ext}^i_{\mathcal{O}_{X_{y'}}}(K_{y'}, E_{y'}) = \mathrm{Ext}^i_{\mathcal{O}_{X_y}}(K_y, E_y) \otimes_k k'$$
by Derived Categories of Schemes, Lemma 0AA7. Thus the vanishing in (1) is really a property of the induced point $y \in |Y|$. We will use these two observations without further mention in the proof.

Assume first Y is an affine scheme. Then we may apply More on Morphisms of Spaces, Lemma 0DKY and find a pseudo-coherent $L \in D(\mathcal{O}_Y)$ which "universally computes" $Rf_*R\mathcal{H}om(K, E)$ in the sense described in that lemma. Unwinding the definitions, we obtain for a point $y \in Y$ the equality
$$\mathrm{Ext}^i_{\kappa(y)}(L \otimes^{\mathbf{L}}_{\mathcal{O}_Y} \kappa(y), \kappa(y)) = \mathrm{Ext}^i_{\mathcal{O}_{X_y}}(K_y, E_y)$$

We conclude that

$$H^i(L \otimes^{\mathbf{L}}_{\mathcal{O}_Y} \kappa(y)) = 0 \text{ for } i > 0 \Leftrightarrow \mathrm{Ext}^i_{\mathcal{O}_{X_y}}(K_y, E_y) = 0 \text{ for } i < 0.$$

By Derived Categories of Schemes, Lemma 0BDI the set W of $y \in Y$ where this happens defines an open of Y. This open W then satisfies the requirement in (1) for all morphisms from spectra of fields, by the "universality" of L.

Let's go back to Y a general algebraic space. Choose an étale covering $\{V_i \to Y\}$ by affine schemes V_i. Then we see that the subset $W \subset |Y|$ pulls back to the corresponding subset $W_i \subset |V_i|$ for X_{V_i}, K_{V_i}, E_{V_i}. By the previous paragraph we find that W_i is open, hence W is open. This proves (1) in general. Moreover, parts (2) and (3) are entirely formulated in terms of the category $Spaces/W$ and the restrictions X_W, K_W, E_W. This reduces us to the case $W = Y$.

Assume $W = Y$. We claim that for any algebraic space V over Y we have $Rf_{V,*}R\mathcal{H}om(K_V, E_V)$ has vanishing cohomology sheaves in degrees < 0. This will prove (2) because

$$\mathrm{Ext}^i_{\mathcal{O}_{X_V}}(K_V, E_V) = H^i(X_V, R\mathcal{H}om(K_V, E_V)) = H^i(V, Rf_{V,*}R\mathcal{H}om(K_V, E_V))$$

by Cohomology on Sites, Lemmas 08JA and 0D6H and the vanishing of the cohomology sheaves implies the cohomology group H^i is zero for $i < 0$ by Derived Categories, Lemma 05TC.

To prove the claim, we may work étale locally on V. In particular, we may assume Y is affine and $W = Y$. Let $L \in D(\mathcal{O}_Y)$ be as in the second paragraph of the proof. For an algebraic space V over Y denote L_V the derived pullback of L to V. (An important feature we will use is that L "works" for all algebraic spaces V over Y and not just affine V.) As $W = Y$ we have $H^i(L) = 0$ for $i > 0$ (use More on Algebra, Lemma 0BCC to go from fibres to stalks). Hence $H^i(L_V) = 0$ for $i > 0$. The property defining L is that

$$Rf_{V,*}R\mathcal{H}om(K_V, E_V) = R\mathcal{H}om(L_V, \mathcal{O}_V)$$

Since L_V sits in degrees ≤ 0, we conclude that $R\mathcal{H}om(L_V, \mathcal{O}_V)$ sits in degrees ≥ 0 thereby proving the claim. This finishes the proof of (2).

Assume $W = Y$ but make no assumptions on the algebraic space Y. Since we have (2), we see from Simplicial Spaces, Lemma 0DL9 that the functor F given by $F(V) = \mathrm{Hom}_{\mathcal{O}_{X_V}}(K_V, E_V)$ is a sheaf[1] on $(Spaces/Y)_{fppf}$. Thus to prove that F is an algebraic space and that $F \to Y$ is affine and of finite presentation, we may work étale locally on Y; see Bootstrap, Lemma 04U0 and Morphisms of Spaces, Lemmas 03WG and 0410. We conclude that it suffices to prove F is an affine algebraic space of finite presentation over Y when Y is an affine scheme. In this case we go back to our pseudo-coherent complex $L \in D(\mathcal{O}_Y)$. Since $H^i(L) = 0$ for $i > 0$, we can represent L by a complex of the form

$$\ldots \to \mathcal{O}_Y^{\oplus m_1} \to \mathcal{O}_Y^{\oplus m_0} \to 0 \to \ldots$$

[1]To check the sheaf property for a covering $\{V_i \to V\}_{i \in I}$ first consider the Čech fppf hypercovering $a : V_\bullet \to V$ with $V_n = \coprod_{i_0 \ldots i_n} V_{i_0} \times_V \ldots \times_V V_{i_n}$ and then set $U_\bullet = V_\bullet \times_{a,V} X_V$. Then $U_\bullet \to X_V$ is an fppf hypercovering to which we may apply Simplicial Spaces, Lemma 0DL9.

with the last term in degree 0, see More on Algebra, Lemma 064U. Combining the two displayed formulas earlier in the proof we find that

$$F(V) = \mathrm{Ker}(\mathrm{Hom}_V(\mathcal{O}_V^{\oplus m_0}, \mathcal{O}_V) \to \mathrm{Hom}_V(\mathcal{O}_V^{\oplus m_1}, \mathcal{O}_V))$$

In other words, there is a fibre product diagram

$$\begin{array}{ccc} F & \longrightarrow & Y \\ \downarrow & & \downarrow 0 \\ \mathbf{A}_Y^{m_0} & \longrightarrow & \mathbf{A}_Y^{m_1} \end{array}$$

which proves what we want. □